



005 Week 1 — English Worksheet (Beginner)

Topic: One Function, Many Coasters — Parameters · **Course:** Theme Park Creator · **Level:** Beginner ·

Time: about 30 minutes

This week you build a roller coaster with a **function**. You give it **parameters** — numbers it uses inside — so the same function can build a long coaster or a short one, just by changing the numbers you **call** it with.

These are the blocks you use this week:

```
rollerCoasterBuilder.add_straight_line(10)
rollerCoasterBuilder.add_ramp(RcbVerticalDirection.UP, 10)
rollerCoasterBuilder.add_ramp(RcbVerticalDirection.DOWN, 10)
```

A direction is **UP** or **DOWN**. The number after it is how many blocks the ramp goes.

1 · Predict 🧠

Read the code. Before you imagine the cart riding it, circle or write your answer.

```
def build_hill(line_len):
    rollerCoasterBuilder.add_straight_line(line_len)
    rollerCoasterBuilder.add_ramp(RcbVerticalDirection.UP, line_len)

build_hill(6)
```

The straight line will be 6 blocks long. Circle one: yes · no

Why?


```
def build_hill(line_len):
    rollerCoasterBuilder.add_straight_line(line_len)
    rollerCoasterBuilder.add_ramp(RcbVerticalDirection.UP, line_len)
```

There is no call after the definition. Does anything get built? Circle one: yes · no

2 · Find the Difference 🐉

Each pair shows clean code first, then a broken version of the same idea. Circle what's different and write one short sentence about the bug.

Pair A — The function should be **called** so the hill gets built.

```
# clean
def build_hill(line_len):
    rollerCoasterBuilder.add_straight_line(line_len)
    rollerCoasterBuilder.add_ramp(RcbVerticalDirection.UP, line_len)

build_hill(6)
```

```
# buggy
def build_hill(line_len):
    rollerCoasterBuilder.add_straight_line(line_len)
    rollerCoasterBuilder.add_ramp(RcbVerticalDirection.UP, line_len)
```

What is wrong? Why does the buggy code build nothing?

Pair B — The function should use the **parameter** `line_len`, not a fixed number.

```
# clean
def build_hill(line_len):
    rollerCoasterBuilder.add_straight_line(line_len)
```

```
# buggy
def build_hill(line_len):
    rollerCoasterBuilder.add_straight_line(10)
```

What is wrong?

3 · Fill the Gap

This coaster should go **up** and then come back **down** so the cart lands safely. One word is missing. Fill it in using the word bank.

```
def up_and_down(ramp_len):  
    rollerCoasterBuilder.add_ramp(RcbVerticalDirection.UP, ramp_len)  
    rollerCoasterBuilder.add_ramp(RcbVerticalDirection.____, ramp_len)  
  
up_and_down(10)
```

Word bank: DOWN · UP · LEFT

Write the missing word:

4 · Tell Me What You Built 📷

Now switch to your homework world. Write one `build_hill` function with a `line_len` parameter and a ramp up, then call it two times with different numbers. When you finish, come back here.

Send a photo or video of your coaster, then explain what you did. Use these sentence starters — write 2 or 3 sentences.

My function's parameter was ...

When I called it with a bigger number, ...

One tricky moment was when ...

5 · Record Your Walkthrough 🎥

Take a video on your phone (or a parent's phone) while your code builds the coaster. Talk like you are teaching a friend. Try to use these words: **function**, **parameter**, **call**.

1. Show your function and run your code.
2. Show two coasters built with different numbers.
3. Say in your own words what a **parameter** is.

Write what you will say in your video. You can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.